

Machine Learning Engineer

Type Assessment

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Content

1.	Introduction and Submission.....	2
2.	Task Description	2
3.	Requirements.....	2
4.	Dataset	3
5.	Evaluation Criteria.....	4

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1. Introduction and Submission

Welcome to the machine learning engineer assessment from the team. **Please confirm the receipt of this task by email to the technical contact.** If you have any questions, please get in touch with the technical contact(s).

Please send your results and files as a zip file via email. Your codes and results should be reproducible, i.e we can run your code on our own local machine. You have **a week (since you received the email)** to complete the task.

The goal of this assessment is to evaluate your ability to design and implement a reproducible, modular, and extendable ML pipeline using a pretrained Transformer model for a text classification task. **We are not evaluating model performance — the focus is on code structure, design clarity, and reproducibility.**

2. Task Description

You are required to build an **end-to-end text classification pipeline** using a pretrained model (we recommend using [distilbert/distilbert-base-german-cased](#)) from the Hugging Face Transformers library. Your solution should consist of at least **two main scripts**:

1. **train_pipeline.py**

This script should:

- Take the dataset as input.
- Perform some preprocessing.
- Fine-tune the pretrained model and track relevant configuration files or hyperparameters for later use in the inference phase.
- Evaluate the fine-tuned model on a test split.
- Save the **best model checkpoint**.
- We recommend using [Mlflow](#) to track hyperparameters, model performance, and to save the model.

2. **inference.py**

This script should:

- Load the **best saved model** from the previous step.
- Accept new text inputs (e.g., as command-line arguments or from a local JSON file or what you think is easier to use).
- Perform the preprocessing steps and prediction using the trained model.
- Return the predicted class/label(s).

3. Requirements

Your solution should demonstrate:

1. Reproducibility

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- include a requirements.txt file or a conda/venv environment file.
 - Code should be runnable from scratch on any machine.
 - Clear documentation in a README.md describing setup, training, and inference steps.
2. Code Structure
 - Modular and well-organized Python code.
 - Separate functions/classes for preprocessing, training, evaluation, and inference.
 - Follow best practices (naming conventions, comments, logging, error handling).
 3. Redeployability
 - Ensure both scripts can be executed independently.
 - Model paths and configurations should be parameterized (not hard-coded).
 - Containerization with **Docker** is a **bonus** — provide a Dockerfile if possible.
 4. Output & Deliverables
 - Source code:
 - train_pipeline.py
 - inference.py
 - all relevant files needed to run the pipeline
 - requirements.txt or environment file
 - (Optional) Dockerfile
 - Short report or presentation that contains the sketch or diagram of your pipeline solution
 - **README.md with clear setup and usage instructions**

4. Dataset

A dataset file was attached to the email (if not, please inform us). The dataset provided contains structured information extracted from real-world text entries describing construction or technical work items. Each record includes both textual and numerical details. The task is to classify each entry into a **label category** based on its description and associated fields.

Column	Type	Description
id	Integer	A unique identifier for each record. Used only for reference; not a predictive feature.
text	String	A text description of a material, service, or task (e.g., product specifications, work activities). This is the main input for your text classification model.
quantity	Numeric	The number of items or hours associated with the entry. Optional feature — may be used if you decide to include numeric embeddings or concatenation with text features.
unit	String	The measurement unit of the quantity (e.g., “Stk”, “Std.”, “m”). Optional feature.
price	Numeric	The price associated with the entry. Optional feature.
label	String	The target variable to be predicted. It indicates whether the entry refers to a material , labor or other item.

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5. Evaluation Criteria

Category	Description	Weight
Code Structure & Clarity	Readability, organization, documentation	30%
Reproducibility	Can others easily reproduce your results?	25%
Pipeline Design	Modularity, reusability, parameterization	25%
Deployment Readiness	Environment setup, Dockerization (bonus), portability	20%